



ST BENEDICT'S

|           |              |
|-----------|--------------|
| SUBJECT   | Chemistry    |
| GRADE     | 11           |
| EXAMINER  | Mrs Govender |
| MODERATOR | Mr Ntombela  |
| DATE      | 23 July 2025 |
| MARKS     | 135          |
| DURATION  | 120 minutes  |

%

LEARNER NAME:

KEY:

D ☐

E ☐

F ☐

TEACHER:

ZN ☐

KH ☐

FMC ☐

**PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY**

1. This question paper consists of 16 pages and a Data Sheet. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment.
4. Remember to write your name and surname in the space provided above AND circle your teachers name.
5. Use the data and formulae sheet whenever necessary.
6. Show your working in all calculations.
7. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
8. Answers must be expressed in decimal format, not left as proper fractions.
9. Where appropriate, express answers to TWO decimal places.
10. It is in your own interest to write legibly and to present your work neatly.
11. Blank pages and blank graphs are included at the end of the paper. If you run out of space for a question, use this space. Clearly indicate the question number of your answer should you use this extra space.

| Question       | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | TOTAL |
|----------------|----|----|----|----|----|----|----|----|-------|
| Mark           |    |    |    |    |    |    |    |    |       |
| Moderated Mark |    |    |    |    |    |    |    |    |       |
| Question Total | 15 | 17 | 18 | 19 | 14 | 17 | 14 | 21 | 135   |

Comment:

Learner's Signature:

Teacher's Signature:

**QUESTION 1      SHARING IS CARING (SOMETIMES)****15 MARKS**

1.1    Intermolecular forces hold compounds together in liquid or solid state.

1.1.1    Explain the formation of London forces between molecules. (4)

---

---

---

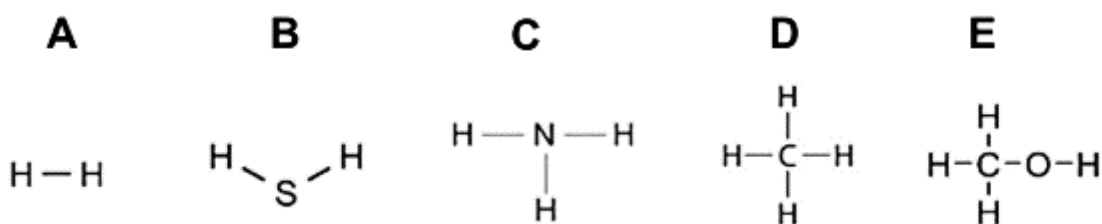
---

---

---

---

Given the following substances labelled A-E and answer the questions that follow.



1.1.2    Substance A has a lower boiling point than substance D. Provide a reason for this occurrence with reference to molecular polarity and intermolecular forces of attraction. (5)

---

---

---

---

---

---

---

1.1.3    Name substance B. (1)

---

1.1.4 Molecule C is polar. Explain why this statement is true. (2)

---

---

---

1.1.5 Draw a diagram illustrating the hydrogen bonding between a single molecule of substance E and water molecules in an aqueous solution. Clearly label ALL possible hydrogen bonds. (3)

**QUESTION 2 TIME TO GET STEAMY**

**17 MARKS**

2.1 Magnesium and oxygen gas react to form magnesium oxide, which has a high melting point.

2.1.1 Write a balanced equation to represent the reaction that takes place. State symbols are not required. (3)

2.1.2 Explain, in terms of its bonding and structure, why magnesium oxide...

a) has a high melting point? (4)

---

---

---

---

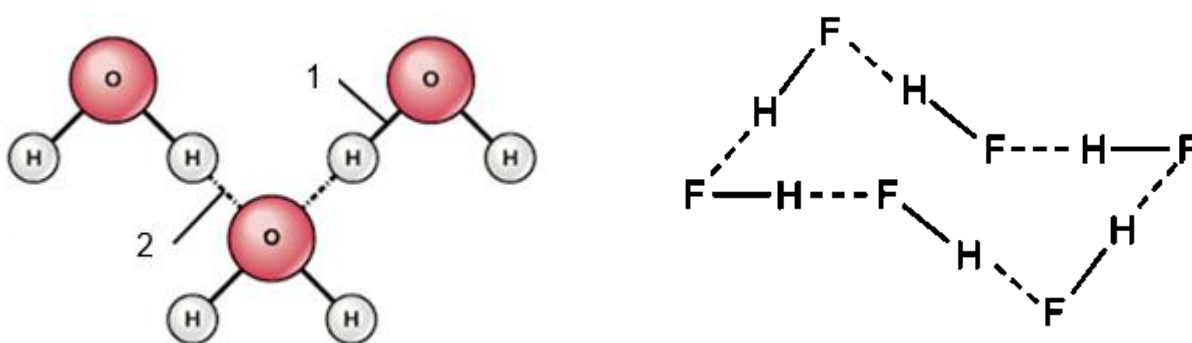
b) does not conduct electricity when solid. (2)

---

---

---

2.2 Consider the diagram below for water and hydrogen fluoride.



2.2.1 Name the specific type of bonding or intermolecular force indicated by 1. (2)

---

---

2.2.2 Why would water have a higher boiling point than hydrogen fluoride, although they both have hydrogen bonds? (2)

---

---

2.3 Methanol has hydrogen bonding between its molecules and iodine has London forces between its molecules, yet the boiling point of methanol is  $64,7^{\circ}\text{C}$  while that of iodine is  $184, 3^{\circ}\text{C}$ .

Account for the difference in the boiling points of methanol and iodine. (4)

---

---

---

---

---

---

---

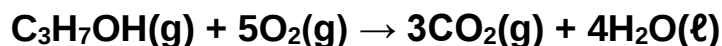
---

---

---

**QUESTION 3      LIMITING REACTANTS-THE PARTY POOPER****18 MARKS**

- 3.1 Propanol burns in oxygen and produces carbon dioxide and water according to the following balanced chemical equation:



- 3.1.1 Define *molar mass*. (1)

---

---

Mika reacts 10 g of propanol with 50 g of oxygen.

- 3.1.2 Use suitable calculations to determine the limiting reactant. (5)

---

---

---

---

---

---

---

- 3.1.3 Determine the total number of oxygen atoms left unreacted. (3)

---

---

---

- 3.1.4 Calculate the volume of CO<sub>2</sub> gas produced at STP. (3)

---

---

---

---

---

- 3.2 When 25 g of impure potassium fluoride is heated, the complete reaction produces 4,48 dm<sup>3</sup> of fluorine gas at STP according to the reaction below:



Calculate the percentage purity of the potassium fluoride. (6)

---

---

---

---

---

---

---

---

---

---

---

---

---

---

---

---

---

---

---

---

---

---

---

---

#### QUESTION 4 THE CHEMICAL COCKTAIL HOUR

19 MARKS

- 4.1 You're mixing two colourful solutions at a chemistry-themed party.

100 cm<sup>3</sup> of 0,4 mol·dm<sup>-3</sup> K<sub>2</sub>SO<sub>4</sub> and 200 cm<sup>3</sup> of 0,3 mol·dm<sup>-3</sup> KBr are mixed together.

- 4.1.1 Write down the dissociation equation for potassium sulphate. (3)

---

---

4.1.2 Determine the oxidation number of sulphur in  $K_2SO_4$ . (2)

---

---

4.1.3 Calculate the total concentration of  $K^+$  ions when these are mixed. (5)

---

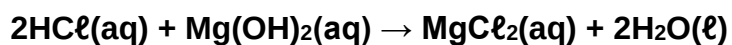
---

---

---

---

4.2 Given the reaction equation between hydrochloric acid and magnesium hydroxide, answer the following questions:



4.2.1 Define a *standard solution*. (2)

---

---

A solution of magnesium hydroxide with concentration  $0,4 \text{ mol.dm}^{-3}$  is required.

4.2.2 Calculate the mass of magnesium hydroxide that should be measured out for 250 ml of solution. (3)

---

---

---

---

---

The steps below indicate the method a student followed to make the standard solution.

- A     *Add pure water to dissolve the salt completely and stir.*
- B     *Rinse the sides of the beaker to ensure salt crystals on the side of the beaker are dissolved.*
- C     *Place empty beaker on the scale and note the reading.*
- D     *Transfer the solution in the beaker to a 250 ml **(Q4.5.1)** flask using a **(Q4.5.2)** to prevent spillage.*
- E     *Fill the flask so that the meniscus is on the calibration mark.*
- F     *Use a spatula to add the correct mass of  $\text{Mg}(\text{OH})_2$  to the beaker.*
- G     *Rinse all solution remaining in the beaker into the flask, seal and shake.*

Refer to step D.

4.2.3    What is the name of the flask used to make the standard solution? (1)

---

4.2.4    Name the apparatus that is used to prevent spillage of the solution. (1)

---

4.2.5    Determine the correct order for the steps shown above and write the letters in the correct sequence from start to finish. (2)

|  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|

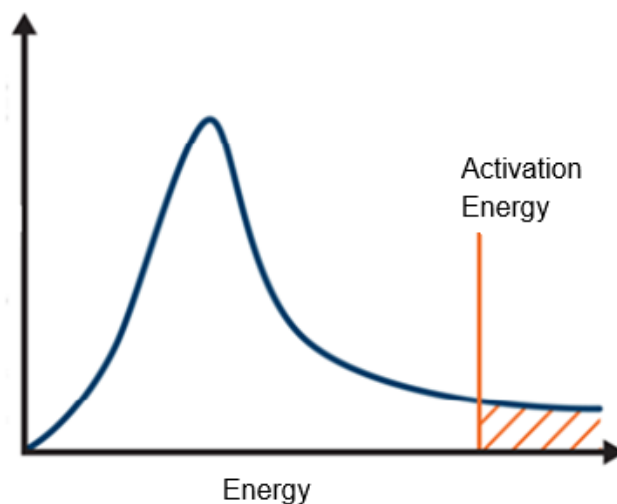
**QUESTION 5      FEELING THE CHILL****14 MARKS**

The breakdown of lactic acid is **exothermic** and has **high activation energy**.

5.1 Draw a fully labelled potential energy diagram for this reaction, showing both the catalysed and uncatalyzed pathways. Include the following labels: (5)

- Activation energy (with and without a catalyst)
- Heat of reaction
- Reactants and products

5.2 Below is a Maxwell-Boltzmann distribution curve for lactic acid molecules in muscle tissue at 37°C. The effect of a catalyst is not yet shown.



5.2.1 What does the area under the curve represent? (2)

---

5.2.2 Indicate on the graph the new activation energy for the catalysed reaction. (2)

5.2.3 "Catalysts speed up the reactions by increasing the energy of molecules. "Do you agree. Circle YES or NO. (1)

Support your answer with reference to theory and the graph. (2)

---

---

---

---

5.2.4 Describe two ways in which the Maxwell-Boltzmann curve shape will change if the muscle temperature is now at 38 °C (2)

---

---

---

#### QUESTION 6 ARE WE THERE YET?

**17 MARKS**

During high intensity exercise, athletes often experience muscle fatigue due to a build up of lactic acid( $C_3H_6O_3$ ). To help with recovery, sport scientists are developing catalyst-based supplements to speed up the breakdown of lactic acid at 37° C. The reaction is:



The table below shows the volume of hydrogen gas released over time during lactic acid breakdown in two trials: one with the catalyst and without.

| Time(s) | Volume of H <sub>2</sub> with catalyst (cm <sup>3</sup> ) | Volume of H <sub>2</sub> without catalyst (cm <sup>3</sup> ) |
|---------|-----------------------------------------------------------|--------------------------------------------------------------|
| 0       | 0                                                         | 0                                                            |
| 10      | 20                                                        | 5                                                            |
| 20      | 35                                                        | 9                                                            |
| 30      | 45                                                        | 12                                                           |
| 40      | 50                                                        | 14                                                           |
| 50      | 50                                                        | 16                                                           |

6.1 Define *reaction rate* in terms of this experiment. (2)

---

---

6.2 Calculate the average rate of H<sub>2</sub> production with the catalyst. (3)

---

---

---

---

---

6.3 Use the data to explain if the catalyst affects:

6.3.1 the amount of product formed (2)

---

---

---

6.3.2 the speed at which the reaction occurs. (2)

---

---

---

6.4 Sketch a graph of volume of H<sub>2</sub> versus time for both trials on the same axes. Include all known values. (4)

- 6.5 Predict the effect if the temperature increased to  $38^{\circ}\text{C}$ , assuming that the catalyst is still functional. Justify your answer using collision theory. (4)

---

---

---

---

---

---

---

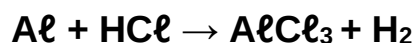
---

---

---

**QUESTION 7      WHEN CHEMISTRY JUST HAPPENS****14 MARKS**

Consider the following unbalanced redox reaction:



7.1 Write the following reactions: (6)

|                         |  |
|-------------------------|--|
| Oxidation half reaction |  |
| Reduction half reaction |  |
| Net ionic reaction      |  |

7.2 Write the formulae of the: (2)

|                       |  |
|-----------------------|--|
| <b>Reducing agent</b> |  |
| <b>Spectator ion</b>  |  |

7.3 Explain why this reaction is spontaneous. (4)

---

---

---

---

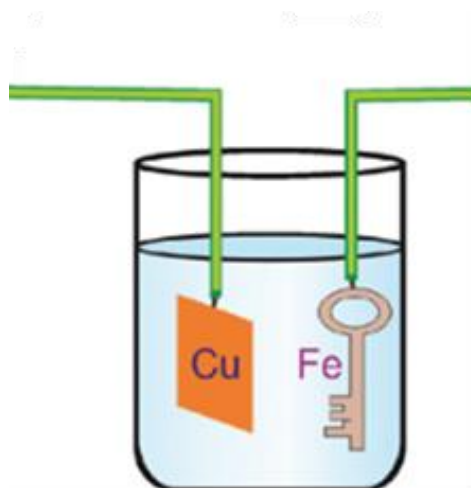
---

7.4 State what happens to a reducing agent during a reaction, in terms of oxidation number and electron transfer. (2)

|                          |  |
|--------------------------|--|
| <b>Oxidation number</b>  |  |
| <b>Electron transfer</b> |  |

**QUESTION 8      WHEN ELECTRICITY FORCES A BREAK UP****21 MARKS**

- 8.1 An application of electrolysis is shown below, in which a key is electroplated with copper.



- 8.1.1 What energy conversion takes place in this electrolytic cell? (2)

- 
- 8.1.2 On the diagram, draw in the battery needed for the cell to work correctly. (2)

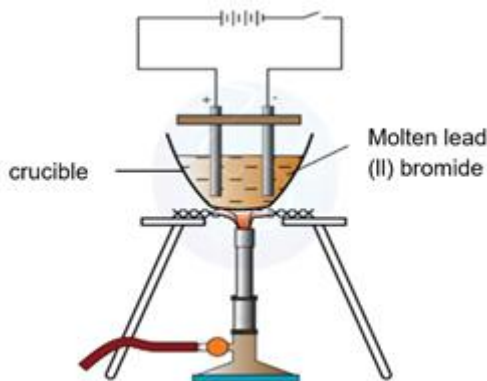
- 8.1.3 Write the chemical formula of a substance that needs to be dissolved in water to form the electrolyte for this electroplating. (2)

- 
- 8.1.4 Write the half reaction that occurs at the key, AND state whether it is the oxidation or reduction half reaction. (3)

---

---

8.2 The apparatus below is used to carry out the electrolysis of molten lead (II) bromide



8.3 Name the material used to make the electrodes in this cell. (2)

8.4 State two reasons why this material is used to make the electrodes (2)

8.5 Why is it necessary to melt the lead (II) bromide? (2)

8.6 What carries electricity in the:

8.6.1 Wire \_\_\_\_\_ (1)

8.6.2 Electrode \_\_\_\_\_ (1)

8.6.3 Molten lead bromide \_\_\_\_\_ (1)

8.7 Complete the table showing details of the electrolysis of molten lead (II) bromide. (3)

|                                      | Negative electrode        | Positive electrode |
|--------------------------------------|---------------------------|--------------------|
| <b>Observations at the electrode</b> | Silvery grey liquid forms |                    |
| <b>Half Reaction</b>                 |                           |                    |

**END OF ASSESSMENT**

### ADDITIONAL SPACE

[illegible]